Afnaan Shafkat Sayyed

Senior Cloud Engineer

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SUMMARY:

- Senior Cloud Engineer with 5+ years of experience designing, deploying and managing scalable, secure and highly available cloud infrastructure across AWS and Azure for healthcare, banking and e-commerce environments.
- Expertise in Infrastructure as Code (Terraform, AWS CloudFormation), CI/CD automation, cloud migration, Kubernetes, networking, monitoring, disaster recovery and cost optimization, with strong hands-on experience in Python, Bash, Jenkins, GitHub Actions and Azure DevOps.
- Skilled in cloud security, IAM, encryption, infrastructure automation, production incident management and root cause analysis, implementing HIPAA-aligned security practices, PCI DSS controls and resilient cloud architectures to improve operational reliability and deployment efficiency.

SKILLS:

Cloud Platforms & Services: AWS, Microsoft Azure, GCP, EC2, S3, VPC, IAM, Lambda, EKS, ECS, Azure VMs, AKS, Azure Functions, Azure Storage, SQL, RDS, DynamoDB, CloudFront, Route 53, CloudWatch, Azure Monitor, KMS, Secrets Manager
Infrastructure as Code & Automation: Terraform, CloudFormation, Azure Bicep, Ansible, Python, Bash, PowerShell, AWS CLI, Azure CLI, Configuration Management, Policy as Code
DevOps & CI/CD: Jenkins, GitHub Actions, GitLab CI/CD, Azure DevOps, Git, Docker, Kubernetes, Helm, Kustomize, Argo CD, GitOps, Automated Deployments, Blue-Green and Canary Releases
Cloud Networking & Architecture: VPC/VNet, TCP/IP, DNS, Subnetting, VPN, Load Balancing, Firewalls, CDN, Private Endpoints, Hybrid Cloud, Multi-Cloud, Microservices, Serverless, Cloud Migration, Landing Zones, High Availability, Multi-Region Architecture, MFA, Centralized Logging
Security, Governance & Compliance: IAM, RBAC, SSO, HashiCorp Vault, Encryption, Vulnerability Scanning, DevSecOps, Zero Trust, AWS Organizations, Azure Policy, HIPAA, HITECH, PCI DSS, SOC 2, NIST, CIS Benchmarks, PHI/ePHI Protection
Monitoring & Site Reliability: Prometheus, Grafana, Splunk, Datadog, ELK Stack, OpenTelemetry, Distributed Tracing, Alerting, Incident Management, Root Cause Analysis, SLOs, SLIs, SLAs, Disaster Recovery, Backup and Recovery, RTO/RPO
Cloud Operations & FinOps: Linux, Windows Server, Nginx, Apache, ITIL, Production Support, Patch Management, Performance Tuning, Capacity Planning, AWS Cost Explorer, Azure Cost Management, Rightsizing, Cost Allocation, Business Continuity
Databases & Integration: PostgreSQL, MySQL, SQL Server, Oracle, Redis, Database Migration, Replication, REST APIs, API Gateway, Message Queues, Event-Driven Architecture
Healthcare, Finance & E-Commerce: HL7, FHIR, EHR/EMR Integration, Healthcare Claims Platforms, Secure Payment Processing, Payment Gateway Integration, Transaction Processing, Order and Inventory Management, Scalable Web Applications, Data Protection, Regulatory Compliance
Tools & Delivery Practices: Jira, Confluence, ServiceNow, PagerDuty, Agile, Scrum, Change Management, Technical Documentation, Runbooks, Architecture Reviews, Disaster Recovery Testing

WORK EXPERIENCE:

Senior Cloud Engineer || UnitedHealth Group, USA **June 2025 – Present**

- Engineered AWS and Azure infrastructure for healthcare applications, claims processing, analytics and AI/ML workloads across development, testing, staging and production environments, improving deployment consistency and scalability.
- Automated provisioning of 100+ cloud resources using Terraform and Python, reducing environment setup time by 50% through reusable modules and standardized configurations across AWS accounts and Azure subscriptions.
- Designed private networking across AWS VPCs and Azure VNets, securing connectivity for 20+ application and data services while preventing direct public access to PHI workloads.
- Deployed containerized healthcare applications and ML inference services on Amazon EKS and Azure Kubernetes Service (AKS), configuring autoscaling, health checks and resource limits to support a 99.9% production availability target.
- Standardized CI/CD pipelines using GitHub Actions and Azure DevOps, integrating automated validation and security scanning to reduce deployment effort by 35% and improve release consistency.
- Implemented healthcare data protection controls using AWS IAM, Microsoft Entra ID and Azure Key Vault, strengthening least-privilege access, encryption and audit logging for sensitive PHI under HIPAA-aligned security practices.
- Provisioned secure infrastructure for healthcare analytics and AI workloads using Amazon S3, Amazon SageMaker, Azure Data Lake Storage Gen2 and Azure Databricks, supporting scalable data processing and model development environments.

- Enabled production AI/ML and Generative AI workloads by supporting secure model endpoints, containerized inference services and API connectivity using Amazon SageMaker and applicable managed AI services, with controlled access to sensitive healthcare data.
- Integrated Amazon CloudWatch and Azure Monitor to track infrastructure health, application performance and operational alerts, reducing mean time to resolve infrastructure incidents by 25% through improved detection and root-cause analysis.
- Implemented AWS Backup and Azure Backup with documented recovery runbooks, targeting a 15-minute recovery-point objective (RPO) and a 4-hour recovery-time objective (RTO) for critical workloads.
- Optimized AWS and Azure resource utilization using AWS Cost Explorer and Azure Cost Management, reducing compute and storage spend by 15% through rightsizing, autoscaling and removal of underutilized resources.

Cloud Engineer || Mphasis, India **July 2022 - Aug 2024**

- Executed phased migration of 50+ banking application servers from on-premises data centers to AWS using AWS Application Migration Service (MGN), EC2, VPC, completing cutovers with validated application connectivity & documented rollback procedures.
- Provisioned standardized cloud infrastructure using Terraform and AWS CloudFormation, reducing manual environment provisioning effort by 35% across development, UAT and production environments.
- Migrated 5+ banking databases using AWS Database Migration Service (DMS), configuring replication tasks and validating data consistency, replication lag, schema compatibility and application connectivity to support reliable production cutovers.
- Enforced least-privilege access through IAM roles and policies, implemented KMS encryption and private subnet isolation and enabled CloudTrail audit logging to improve access governance and traceability of infrastructure changes.
- Automated deployment workflows using Jenkins, Git and AWS deployment services, introducing validation checks, change approvals and rollback procedures to improve release consistency and reduce manual deployment effort.
- Configured CloudWatch dashboards, alarms and centralized logs to monitor CPU utilization, memory pressure and infrastructure errors, reducing mean time to resolution (MTTR) by 25% through faster incident identification and troubleshooting.
- Reduced monthly AWS infrastructure expenditure by 15% through EC2 rightsizing, removal of idle resources and S3 storage lifecycle policies while maintaining required workload performance.
- Developed Python and Bash scripts to analyze CloudWatch metrics, flag sustained CPU and memory spikes and identify recurring resource bottlenecks, enabling proactive capacity adjustments and more consistent infrastructure health checks.
- Maintained disaster recovery readiness by validating EBS snapshots and AWS Backup jobs, documenting recovery procedures and participating in failover tests against application-specific RTO/RPO requirements.

Cloud Engineer || Infosys, India **Oct 2020 - June 2022**

- Provisioned and managed AWS infrastructure using EC2, VPC, subnets, security groups, Application Load Balancer and Auto Scaling to host scalable e-commerce workloads and handle seasonal traffic fluctuations.
- Automated infrastructure provisioning with Terraform and AWS CloudFormation across Development, QA and Production environments, reducing manual setup effort by 30% and improving configuration consistency.
- Engineered CI/CD pipelines using Jenkins, Git and Maven to automate application builds, deployments and release validation, reducing deployment turnaround time by 25%.
- Configured AWS S3, RDS, IAM and Systems Manager to protect application data, manage environment configurations, enforce least-privilege access and improve deployment consistency.
- Implemented multi-AZ architecture, load balancing, Route 53 routing and backup and recovery procedures to strengthen application availability and business continuity during infrastructure failures.
- Established CloudWatch dashboards, alarms and CloudTrail audit trails, applying Linux troubleshooting and log analysis to identify infrastructure bottlenecks and reduce initial incident diagnosis time by 20%.
- Enforced cloud security through IAM policies, network ACLs, security groups, encryption and OS patching; maintained audit evidence for applicable PCI DSS controls in payment-card environments.
- Resolved production infrastructure incidents and performed root cause analysis, documenting corrective actions in operational runbooks to help prevent recurring issues and reduce repeat escalations.

EDUCATION:

Master's in Information Systems - DePaul University**, IL, USA**
Bachelor of Engineering in Mechanical Engineering - Vishwakarma Institute of Information Technology**, India**

CERTIFICATIONS:

Microsoft Certified: **DevOps Engineer Expert (AZ-400)**
Microsoft Certified: **Azure Developer Associate (AZ-204)**
Microsoft Certified: **Identity and Access Administrator Associate (SC-300)**
Microsoft Certified: **Information Protection and Compliance Administrator Associate (SC-400)**